MUSCLE GUARD
RESEARCH HUB · EVIDENCE SUMMARY

GLP-1 + Muscle: the evidence in 12 pages.

A plain-English summary of what the research says about GLP-1 weight loss and lean mass. With sources.

Compiled by the Muscle Guard research team.
Last updated 6 June 2026 · Free to share.
muscleguardglp.com/evidence.pdf

The three-sentence summary.

1. GLP-1 receptor agonists produce significant weight loss — and, without intervention, between roughly a quarter and forty-five percent of what is lost can be lean muscle, not fat. [1,2,3]

2. With adequate protein (1.2 to 1.6 g per kg of body weight per day) and resistance training (2 to 4 sessions per week), that lean-mass-loss share drops substantially. The magnitude of the drop is extrapolated from adjacent (non-GLP-1) intervention trials; a GLP-1-specific intervention trial reporting an exact figure is pending. [4,5,6]

3. Muscle Guard is the GLP-1 companion app that tracks the playbook — protein, training, weight trend, body composition, and a single score that captures fat-loss-without-muscle-loss — across every approved GLP-1, including compounded preparations.

How to read this document.

This is the 12-page version of the Muscle Guard Research Hub at muscleguardglp.com/research/. It is free to download, free to share, and free to send to your healthcare provider. Every numbered claim links to a citation on page 10.

What GLP-1s do to your body.

GLP-1 receptor agonists were originally developed for type 2 diabetes. They suppress appetite, slow gastric emptying, and improve insulin sensitivity. The weight-loss effect is downstream of the appetite suppression — most users eat 30 to 50 percent less than they did before.

The drugs in this class.

Semaglutide (Ozempic, Wegovy, Rybelsus oral) and **tirzepatide** (Mounjaro, Zepbound) are the two molecules that account for the bulk of 2026 prescriptions. **Liraglutide** (Saxenda, daily injection) remains in use. Compounded semaglutide and tirzepatide are widely prescribed in markets where the branded products are inaccessible.

Average weight loss across trials.

Phase-3 head-to-head data: semaglutide at the higher Wegovy dose produces ~14-17% of starting body weight over 68 weeks (STEP-1). [1] Tirzepatide at 15 mg produces ~18-22% over 72 weeks (SURMOUNT-1). [2] Individual variation is wide.

Why lean mass loss happens.

A sustained calorie and protein deficit forces the body to mobilise both fat and muscle for energy. Without an explicit training stimulus and an adequate protein floor, the body has no reason to preserve muscle. The lean-mass-loss share figures in the major trials are body-composition substudies of those trials, where participants received no specific intervention. [3]

Side effects, week by week.

The side-effect profile is broadly similar across all GLP-1s. The pattern matters more than any single symptom — what is common at week 4 is different from what is common at week 16.

Weeks 1-4 — starting dose.

- Mild to moderate nausea, especially day 1-2 after the injection
- Fatigue, particularly afternoon dips
- Dramatic appetite suppression by week 2
- Constipation, mitigated by hydration and fibre

Weeks 5-12 — titration phase.

- Nausea returns at each step-up dose
- "Sulphur burps" reported by ~10-15% of users at higher doses
- First visible weight loss — typically 4-8% of starting weight
- Sleep disruption in some users

Weeks 13-24 — approaching peak dose.

- GI symptoms generally easier than during escalation
- Plateaus possible — weight loss is rarely linear
- "Food blankness" — nothing sounds appealing
- Muscle loss becomes visible if not actively prevented

When to call your healthcare provider.

Severe abdominal pain, persistent vomiting, signs of dehydration, vision changes, or persistent fatigue. These warrant a clinical conversation, not a discontinuation decision made alone.

Muscle Guard is a self-tracking companion and coach. Not a medical device. Not medical advice.

Comparing the GLP-1s.

Drug	Active	Dosing	Avg fat loss	SA price/mo
Ozempic	Semaglutide	Weekly injection	~10-15%	R2,700-R3,100
Wegovy	Semaglutide	Weekly injection	~14-17%	Not registered
Mounjaro	Tirzepatide	Weekly injection	~18-22%	R3,000-R6,000
Rybelsus	Oral semaglutide	Daily tablet	~5-10%	R2,400-R2,800
Compounded	Semaglutide	Weekly injection	~12-15%	R1,000-R2,500

Efficacy from published phase-3 trial data. SA pricing Q1 2026.

Muscle preservation across the class.

All three branded GLP-1s drive some lean-mass loss in the absence of intervention. Tirzepatide produces larger weight loss in absolute terms, so the absolute lean-mass loss is also larger even if the fraction is similar or smaller. [3] The choice of drug is less important for muscle preservation than your daily habits.

How to protect your muscle — protein.

Lean mass loss on a GLP-1 is largely under your control. Three levers — adequate protein, a resistance-training stimulus, and avoiding extreme calorie deficits — substantially reduce the lean-mass-loss share. The magnitude of the reduction in a GLP-1 context is extrapolated from adjacent intervention trials; a GLP-1-specific intervention trial is pending.

The protein target: 1.2 to 1.6 g/kg/day.

Evidence-based target for adults on a GLP-1: 1.2 to 1.6 grams per kilogram of body weight per day. [4,5] Lower end if sedentary; upper end if resistance-training.

- 60 kg adult — 72 to 96 g protein per day
- 75 kg adult — 90 to 120 g protein per day
- 90 kg adult — 108 to 144 g protein per day

Per-meal target. Spread protein across three or four meals containing 25 to 40 g each. Skipping breakfast — a real risk on a GLP-1 — cuts your protein opportunities meaningfully.

Practical sources: 2 eggs (12 g), 100 g chicken breast (31 g), 100 g salmon (25 g), 200 g Greek yoghurt (20 g), 1 scoop whey protein (24 g), 100 g cottage cheese (11 g), 100 g lentils (9 g), 100 g firm tofu (13 g), 1 tin tuna (25 g).

How to protect your muscle — training.

Resistance training is the non-negotiable second lever. Aerobic exercise is welcome for cardiovascular health but does not by itself preserve muscle in a calorie deficit. [6,7]

The minimum effective dose.

- Frequency: 2 to 4 sessions per week
- Volume: 10 to 20 hard sets per muscle group per week
- Intensity: RPE 7 to 9 (two to three reps shy of failure on most sets)
- Progression: add weight, reps, or sets every one to two weeks

Equipment matters less than consistency.

Bodyweight, dumbbells, resistance bands, or a barbell can all preserve muscle, provided you progressively overload. The published intervention arms used a mix of all four.

What 'good' looks like at 12 weeks.

Users following the protein-and-training plan typically show: weight down 8-12 percent; waist circumference down 5-10 cm; body-fat estimate down 4-6 percentage points; strength on baseline lifts holding or improving; lean mass estimate within 2-3 percent of baseline.

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Body composition tracking.

Weight on a scale is the worst single metric for tracking a GLP-1 journey. It conflates fat loss, lean mass change, and water shifts. Three measurements together — weight, waist circumference, and a body-composition estimate — give a much clearer picture.

Cadence.

- Weight: weekly
- Waist circumference: weekly, same time of day, same posture
- Body composition (DEXA or BIA): every 12 weeks while losing
- Progress photos: monthly, same angles, same lighting

What the Muscle Guard Score captures.

The Muscle Guard Score combines protein adherence (35%), resistance training frequency (25%), weight trend (20%), and body composition (20%) into a 0-100 score. The goal is consistency in the 70+ range across your journey, not chasing 100.

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Maintenance and what comes next.

Most users who stop a GLP-1 regain a meaningful fraction of the lost weight within a year. [8] The fraction is significantly smaller if protein and resistance training continued during the medication phase. The maintenance plan after a GLP-1 is the same plan you should have been running during it — with appetite operating under its own rules again.

What the STEP-1 extension showed.

Participants regained roughly two-thirds of the lost weight in the 12 months following semaglutide discontinuation. Net weight loss shrank from 17.3% on-treatment to 5.6% one year off. [8]

Muscle drain and 'Ozempic face'.

User-coined terms for two related phenomena: the gaunt facial appearance from rapid fat loss combined with lean tissue loss, and the broader sense of fatigue and weakness from losing muscle along with the fat. Both are largely preventable with the protein-and-training plan.

Sources.

Every numbered claim maps to a peer-reviewed source below. Links are clickable.

[1] Wilding JPH et al. (2021). *Once-Weekly Semaglutide in Adults with Overweight or Obesity (STEP-1)*. New England Journal of Medicine 384:989-1002. doi:10.1056/NEJMoa2032183

[2] Jastreboff AM et al. (2022). *Tirzepatide Once Weekly for the Treatment of Obesity (SURMOUNT-1)*. NEJM 387:205-216. doi:10.1056/NEJMoa2206038

[3] Neeland IJ, Linge J, Birkenfeld AL (2024). *Changes in lean body mass with glucagon-like peptide-1-based therapies and mitigation strategies*. Diabetes, Obesity and Metabolism 26(S4):16-27. doi:10.1111/dom.15728

[4] Nunes EA et al. (2022). *Systematic review and meta-analysis of protein intake to support muscle mass and function in healthy adults*. J Cachexia Sarcopenia Muscle 13(2):795-810. doi:10.1002/jcsm.12922

[5] Cava E, Yeat NC, Mittendorfer B (2017). *Preserving Healthy Muscle during Weight Loss*. Advances in Nutrition 8(3):511-519. doi:10.3945/an.116.014506

[6] Longland TM et al. (2016). *Higher compared with lower dietary protein during an energy deficit combined with intense exercise promotes greater lean mass gain and fat mass loss*. American Journal of Clinical Nutrition 103(3):738-746. doi:10.3945/ajcn.115.119339

[7] Murphy C, Koehler K (2022). *Energy deficiency impairs resistance training gains in lean mass but not strength: A meta-analysis and meta-regression*. Scand J Med Sci Sports 32(1):125-137. doi:10.1111/sms.14075

[8] Wilding JPH et al. (2022). *Weight regain and cardiometabolic effects after withdrawal of semaglutide: The STEP 1 trial extension*. Diabetes, Obesity and Metabolism 24(8):1553-1564. doi:10.1111/dom.14725

Methods note.

About the 25-40% claim.

The un-intervened lean-mass-loss share figure of 25-40 percent is taken from published DEXA substudies of STEP-1 (Wilding 2021, Lebovitz 2021 exploratory analysis) and SURMOUNT-1 (Jastreboff 2022), consolidated in Neeland et al. (2024). [1,2,3]

About the 'drops substantially' claim.

The directional claim that adequate protein and resistance training substantially reduce the lean-mass-loss share is supported by adjacent (non-GLP-1) intervention trials: Longland 2016 demonstrated lean-mass gain with high protein and concurrent training during a 40% energy deficit; Murphy & Koehler 2022 meta-analysis showed that an energy deficit blunts but does not eliminate resistance-training-driven lean-mass gains; the Nunes 2022 meta-analysis pooled protein-intake trials in healthy adults. [4,6,7]

What is not yet published. A trial that puts GLP-1 users on the protein-and-training intervention, measures lean-mass-loss share by DEXA, and reports the resulting figure has not yet appeared in the literature as of June 2026. When it does, the directional language in this document will be replaced with the cited figure.

Found something wrong?

Email research@muscleguardglp.com. We update this PDF when corrections are warranted and bump the version date in the footer.

About Muscle Guard.

Muscle Guard is the GLP-1 companion app built around muscle preservation. The only one in the category that puts the question — "what are you keeping of yourself?" — on the front page.

The app tracks injections and tablets across every approved GLP-1, scans plates with AI for protein and macros, monitors body composition, and produces a one-page summary your healthcare provider can read in 60 seconds. Free tier forever. Pro tier R99/month in South Africa, $5.99 in the US, €5.99 in the EU. Seven-day free trial. No card needed.

Built in Johannesburg by HK Brand Expert (Pty) Ltd. POPIA, GDPR and CCPA compliant by architecture. No third-party trackers. Delete in one tap.

Boundary.

Muscle Guard is a self-tracking companion and coach. Not a medical device. Not medical advice. Not affiliated with any prescriber, telehealth provider, or pharmaceutical manufacturer. Always consult your healthcare provider for personal decisions.

Where to read more.

- muscleguardglp.com/research/ — the full Research Hub, 14 articles
- muscleguardglp.com/research-library.pdf — the 30-page comprehensive library
- muscleguardglp.com/story.html — why Helen built Muscle Guard

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